

Trustworthy ML

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Attendance
387201

Vevox



Learning Outcomes

By the end of this session you will be able to:

- Reflect on practical ways of ensuring safety
- Reflect on limitations of measuring model accuracy
- Recognise the relationship between various quality metrics

AI risk must be treated as seriously as climate crisis, says Google DeepMind chief

Demis Hassabis calls for greater regulation to quell existential fears over tech with above-human levels of intelligence

● [Hope or horror? The great AI debate dividing its pioneers](#)

British spy agencies warn of AI threat to democracy

Humanity could 'lose control' of AI, UK PM warns, as Britain seeks leading role in the tech

Britain's Prime Minister Rishi Sunak on Thursday warned of the risks of artificial intelligence ahead of a key summit next week on the...

Human extinction risk from AI on same scale as pandemics or nuclear war, Sunak warns

A.I. Poses 'Risk of Extinction,' Industry Leaders Warn

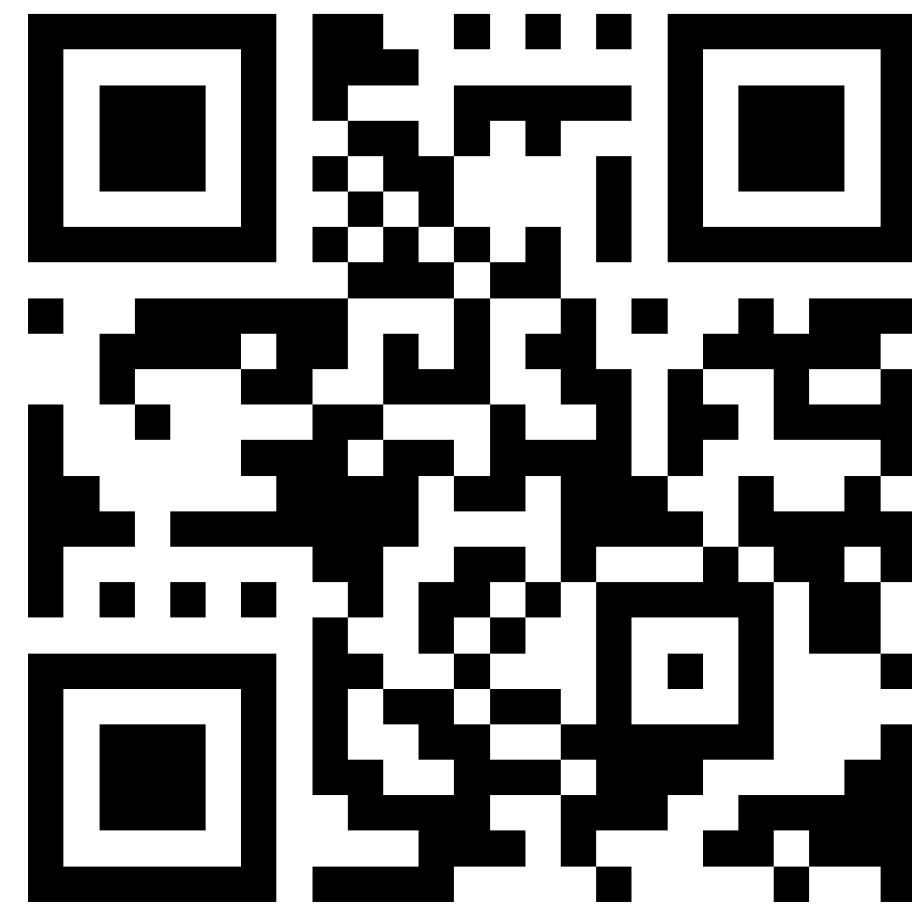
Leaders from OpenAI, Google DeepMind, Anthropic and other A.I. labs warn that future systems could be as deadly as pandemics and nuclear weapons.

Biden to Use Emergency Powers to Mitigate AI Risks

Executive order seeks to manage artificial-intelligence threats from privacy to national security

Google Brain cofounder says Big Tech companies are lying about the risks of AI wiping out humanity because they want to dominate the market

Would you trust a machine learning model with your medical diagnosis?



Some evidence for

Liu, Xiaoxuan, et al. "A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis." *The lancet digital health* 1.6 (2019): e271-e297.

What would you ask the ML developers for you
to feel more confident in the model they
trained?

- What type of model did you use?

- What type of model did you use?
- How many training and test points did you have?

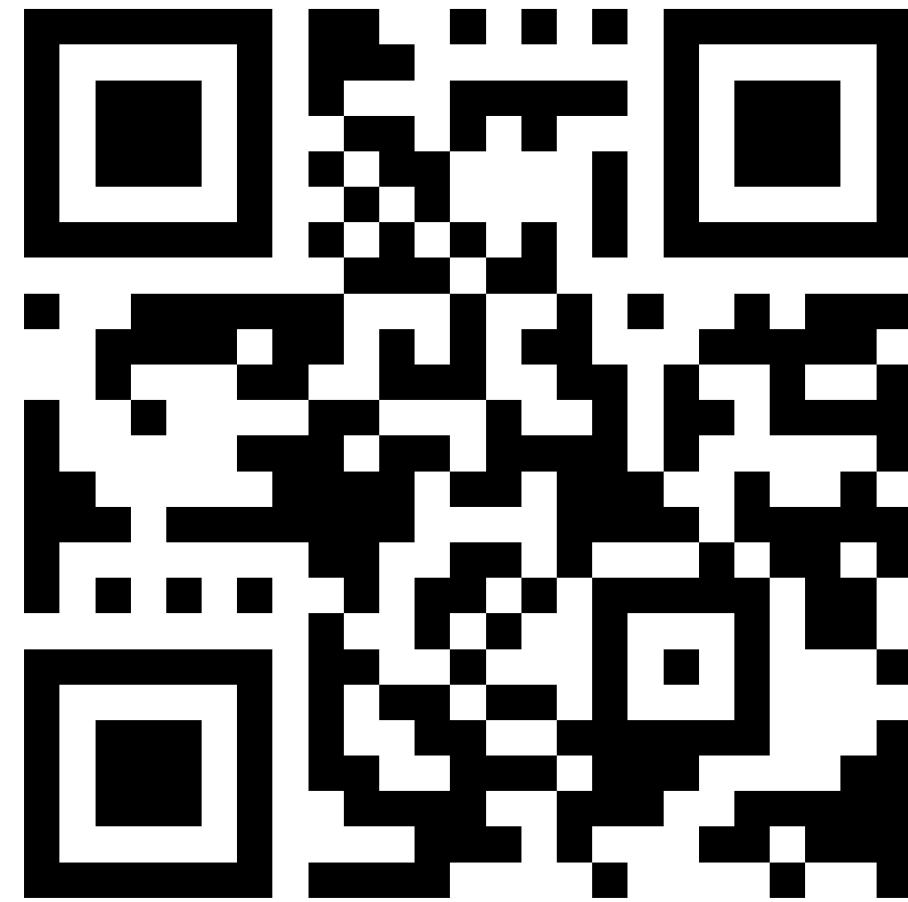
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- How many samples did you have for each class?

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- How many training and test points did you have?
- How many samples did you have for each class?
- What is the accuracy of your model?

When exactly can we say a model
generalises?

Generalisation

Quick Recap



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Not so clear

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- Comparable to human accuracy?
- Sometimes machines do better. How much better?

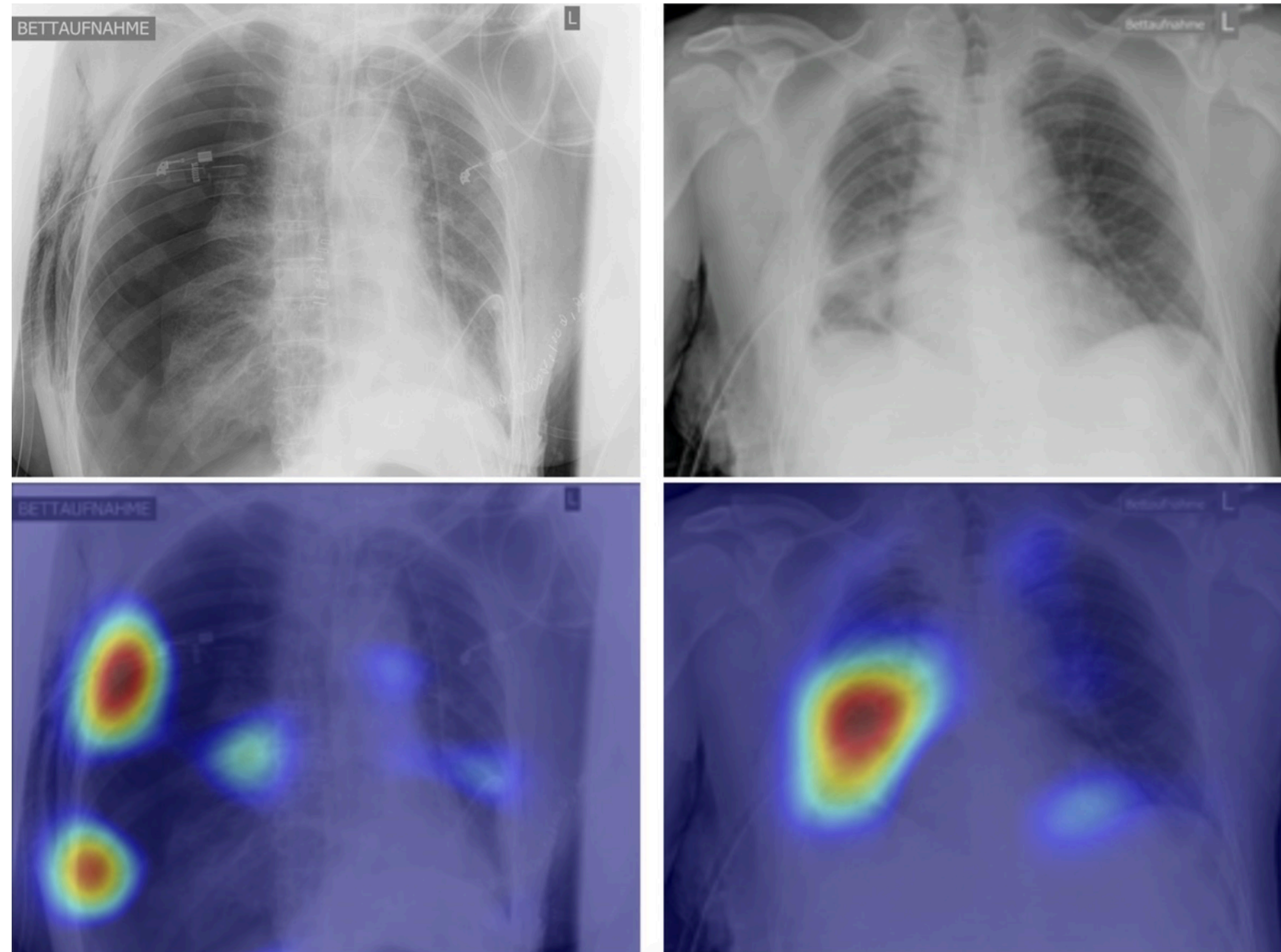
When exactly can we say a model generalises?

Not so clear

- Comparable to human accuracy?
- Sometimes machines do better. How much better?
- Problematic (think e.g. regulations)

Let's assume we have a clear
notion of "good enough" accuracy

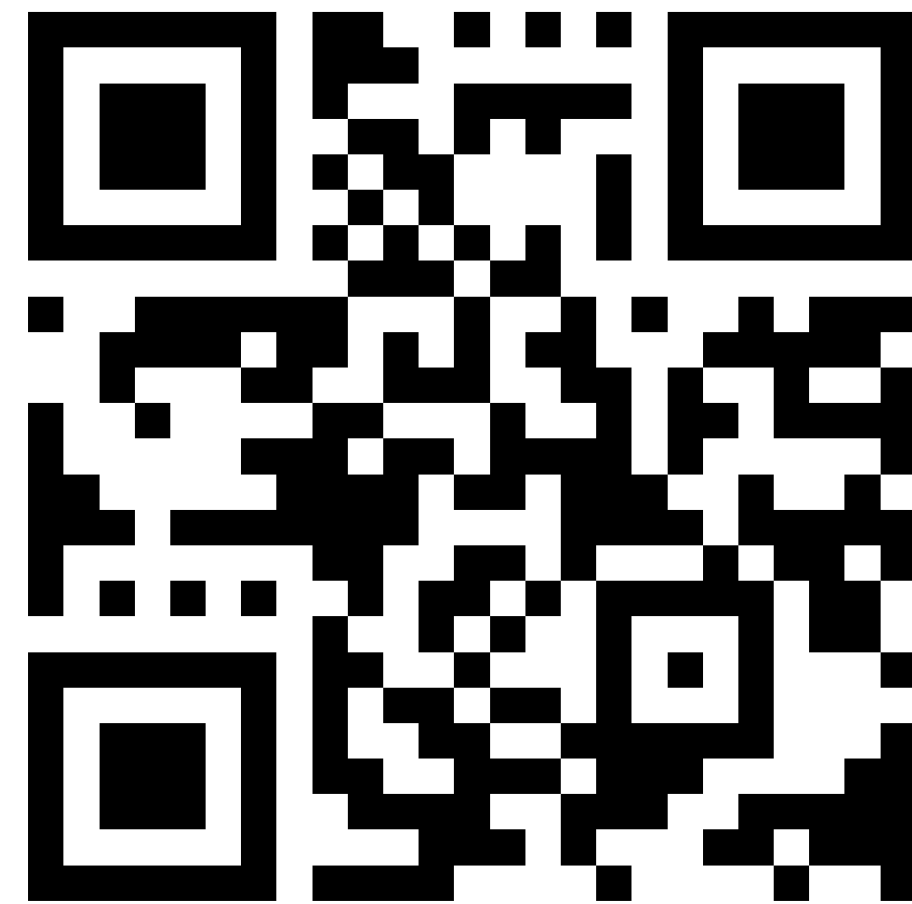
Why Accuracy is Not All We Should Care About



Spurious Correlations



Reflect: How would you check that your model is not relying on spurious correlations?



Practical Stuff You Could Consider

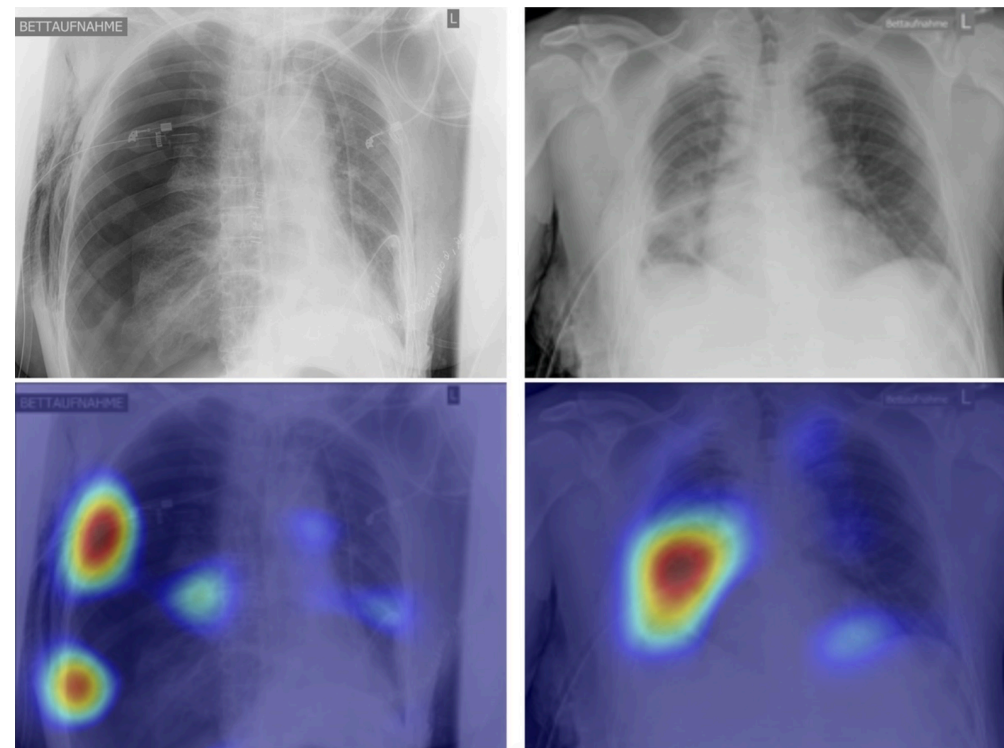
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 1. Difficult to know what you don't know

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 2. Confounding factors

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- Get some other entity/institution to test it out for you
- Analyse what your model has learned
 - Visually: methods can fail (have lots of limitations)
 - Experimentally:
 1. Difficult to know what you don't know
 2. Confounding factors
- Can we do something more principled about it?

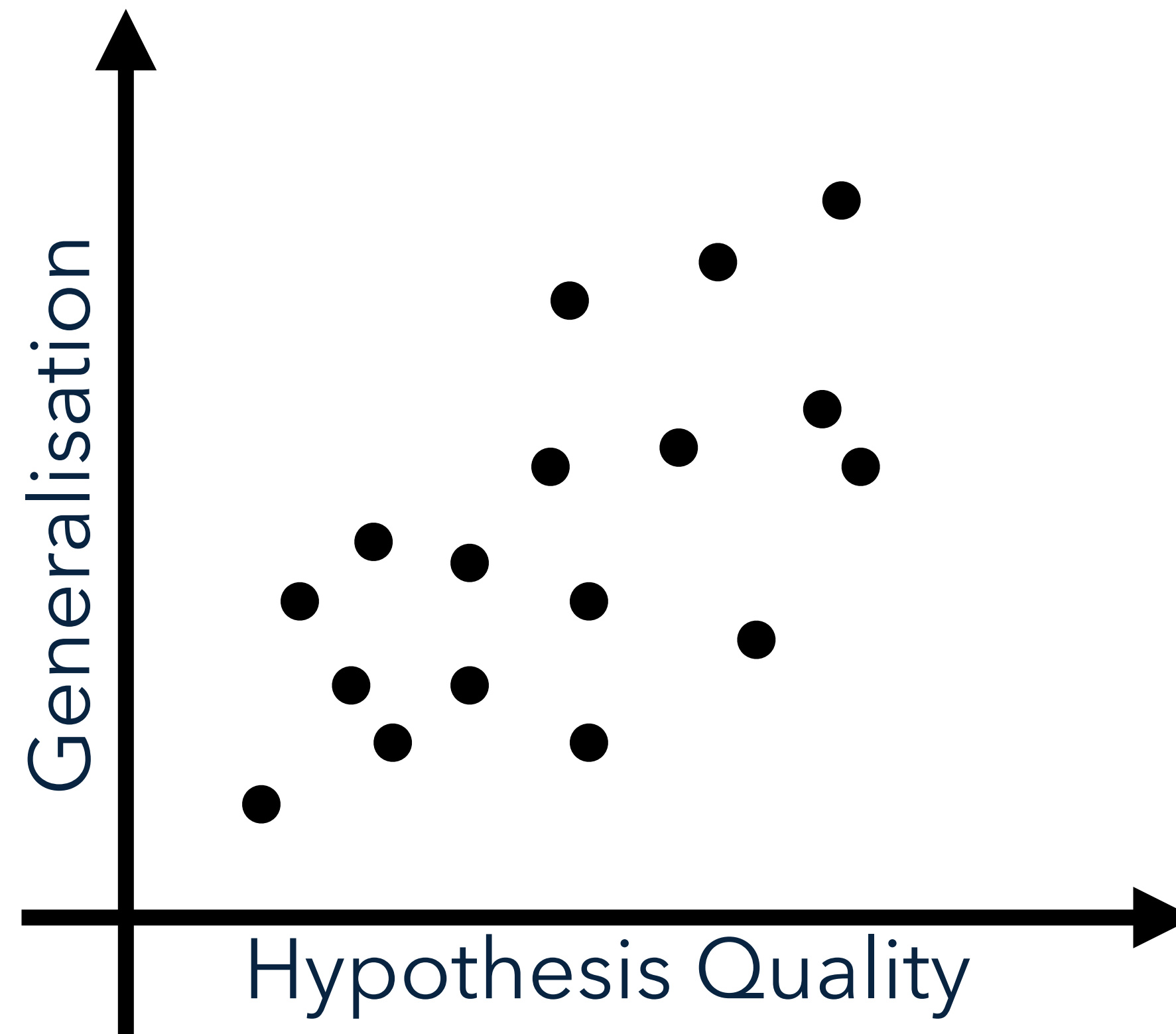
Generalisation Theory

Rigorous

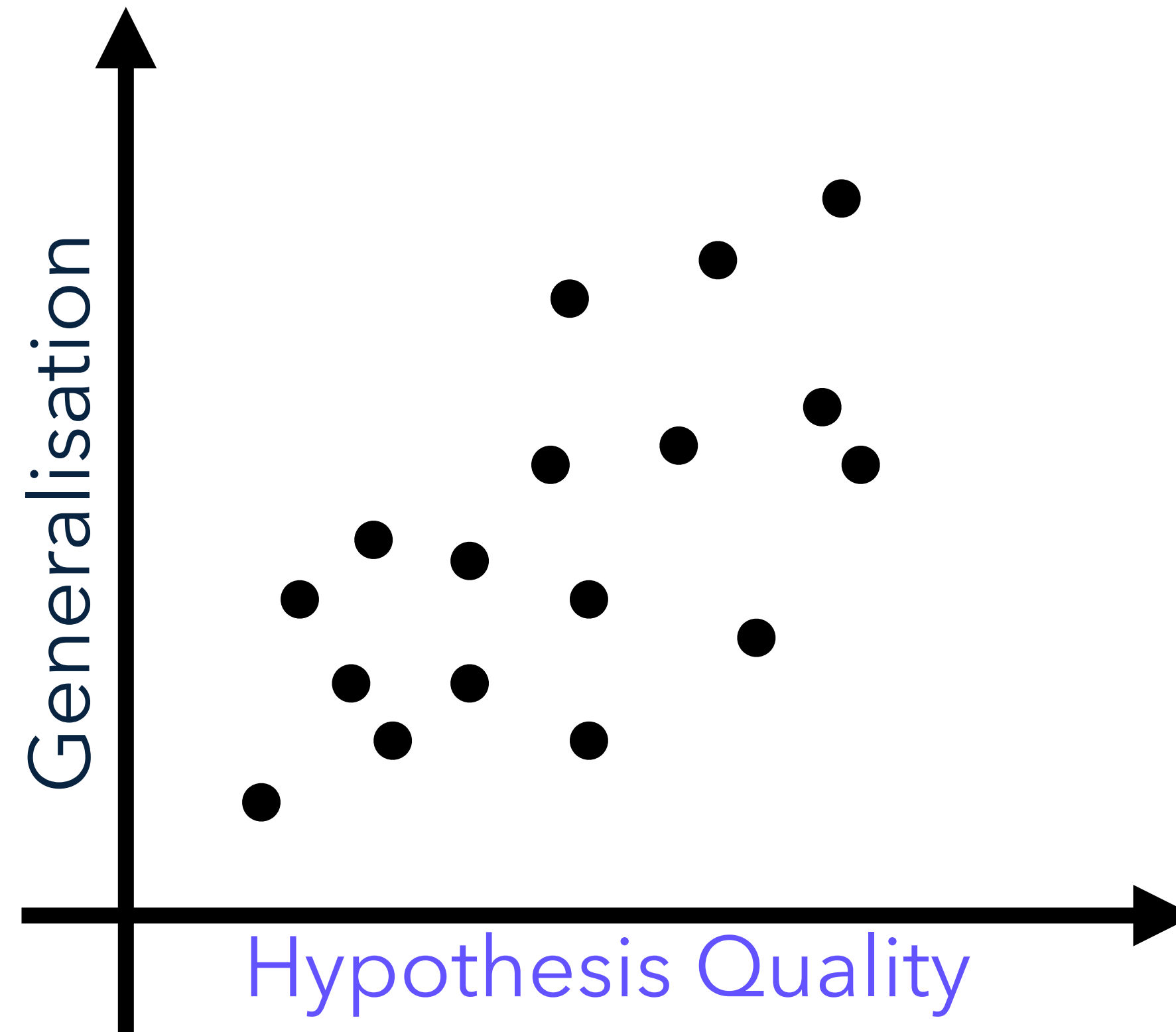
Impractical

THE NEW Generalisation Theory

Objective: Predict Generalisation



Objective: Predict Generalisation



Qualities of a Model?

Model Qualities

- Today we are discussing:
 1. Calibration
 2. Robustness
 3. Expressivity

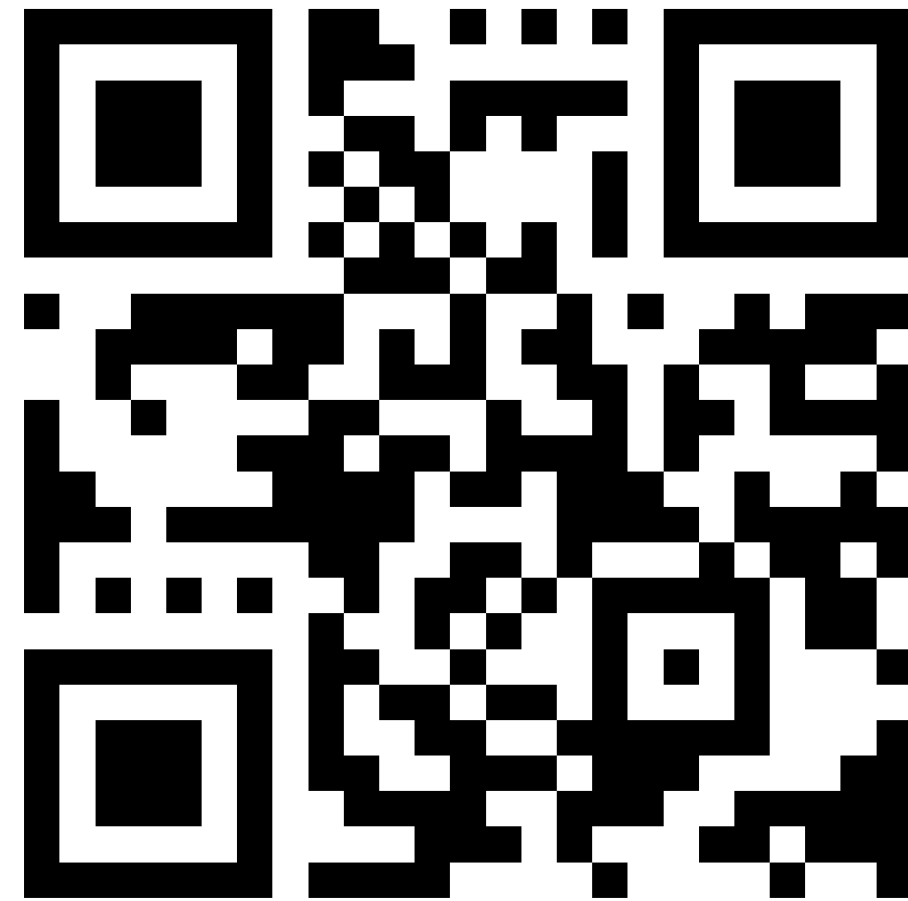
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1. Calibration

Which one is the most dangerous to predict?

- True positive, True negative, False positive, False negative

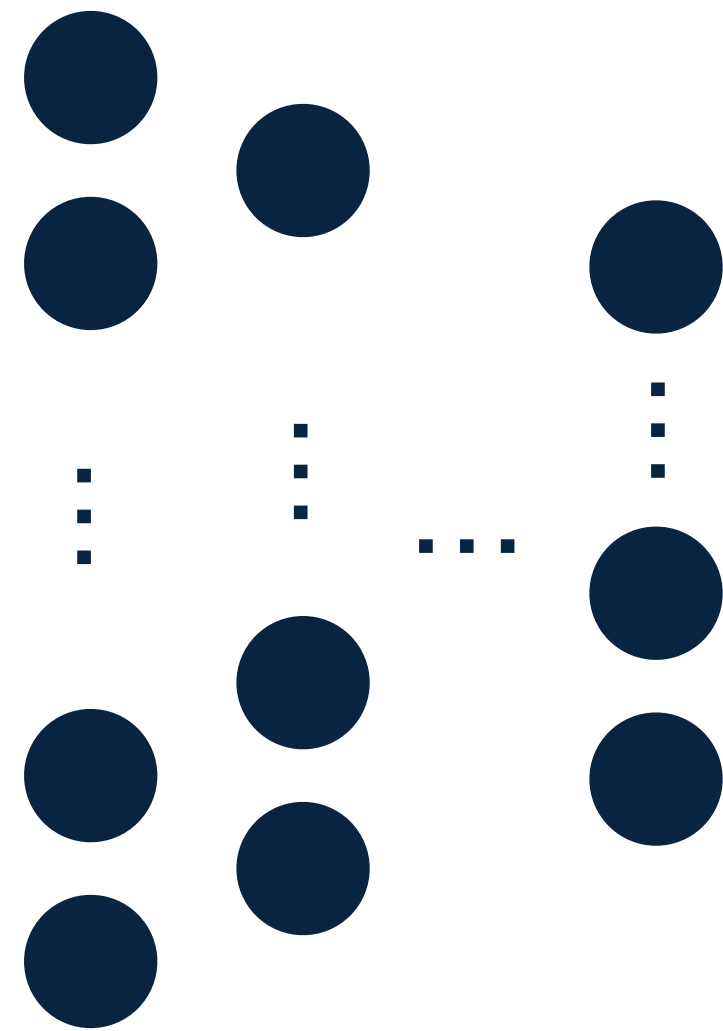


How can we make sure we don't
have false negatives?

Outputs as Probabilities

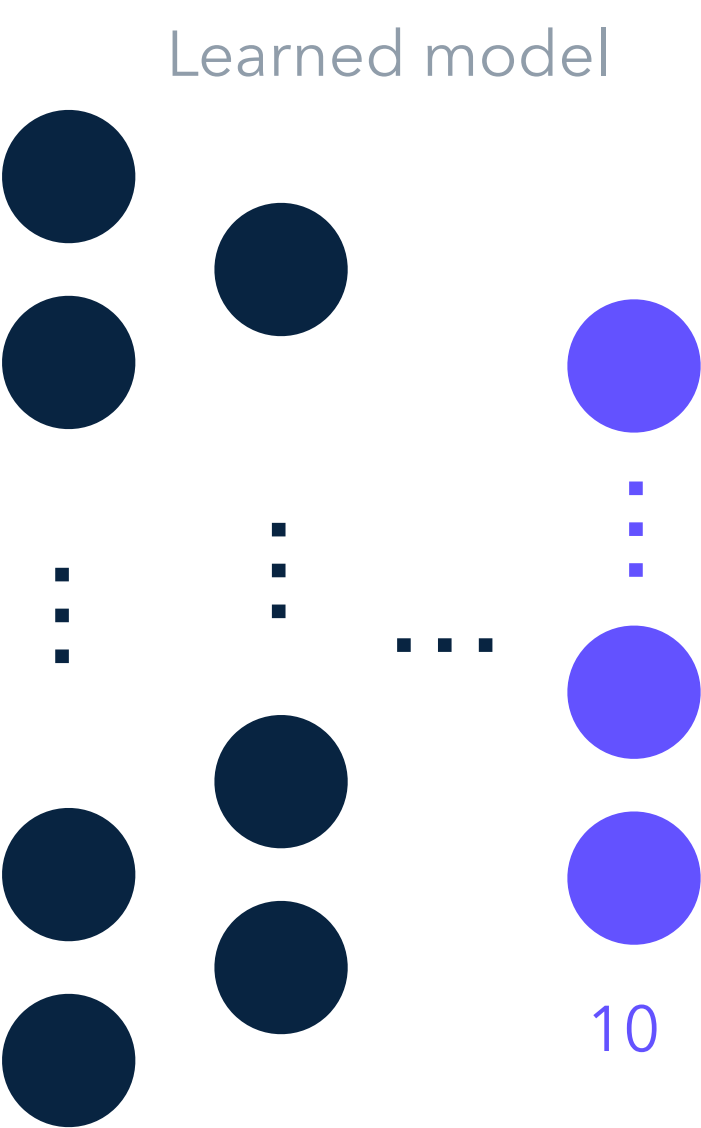
Quick Recap

Learned model



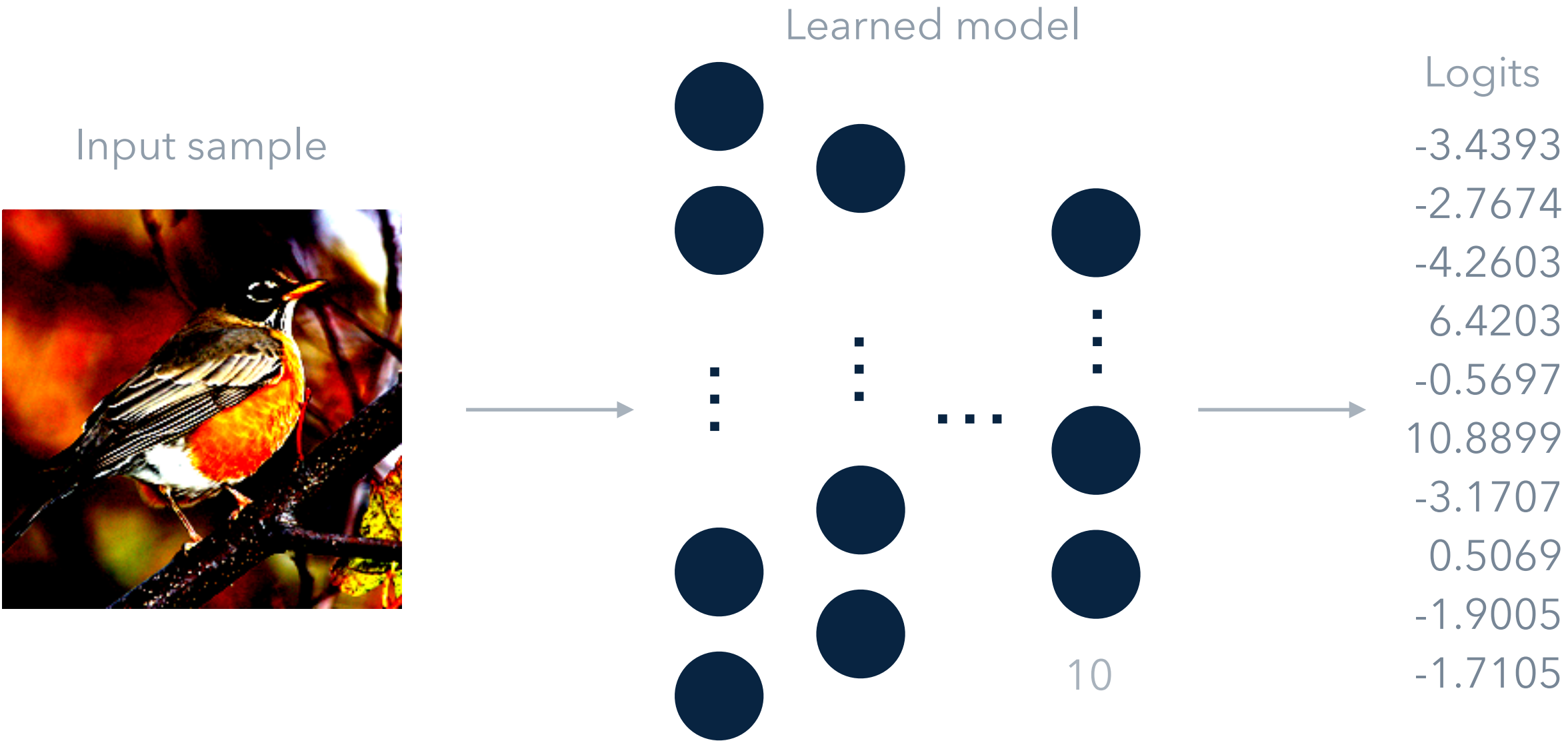
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Quick Recap



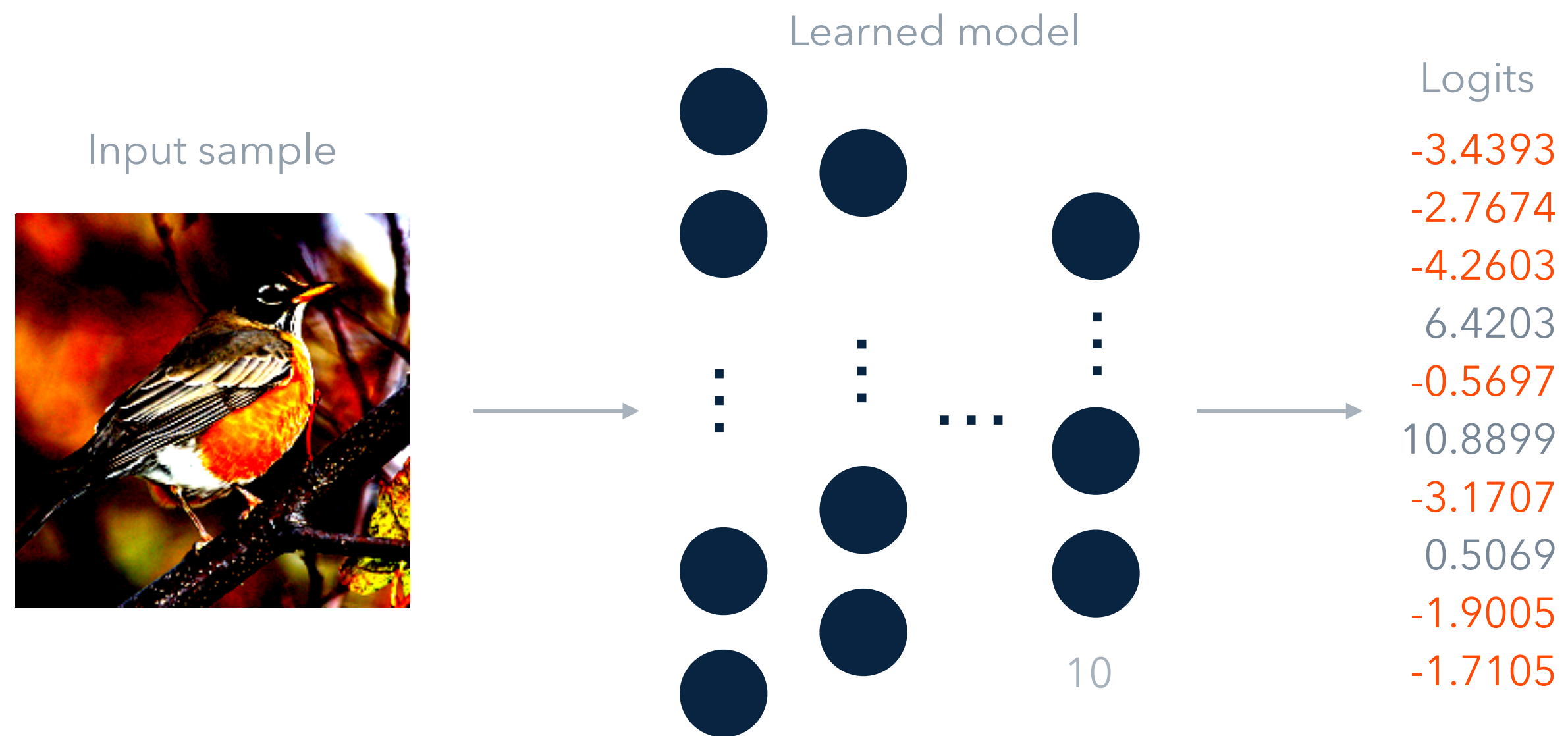
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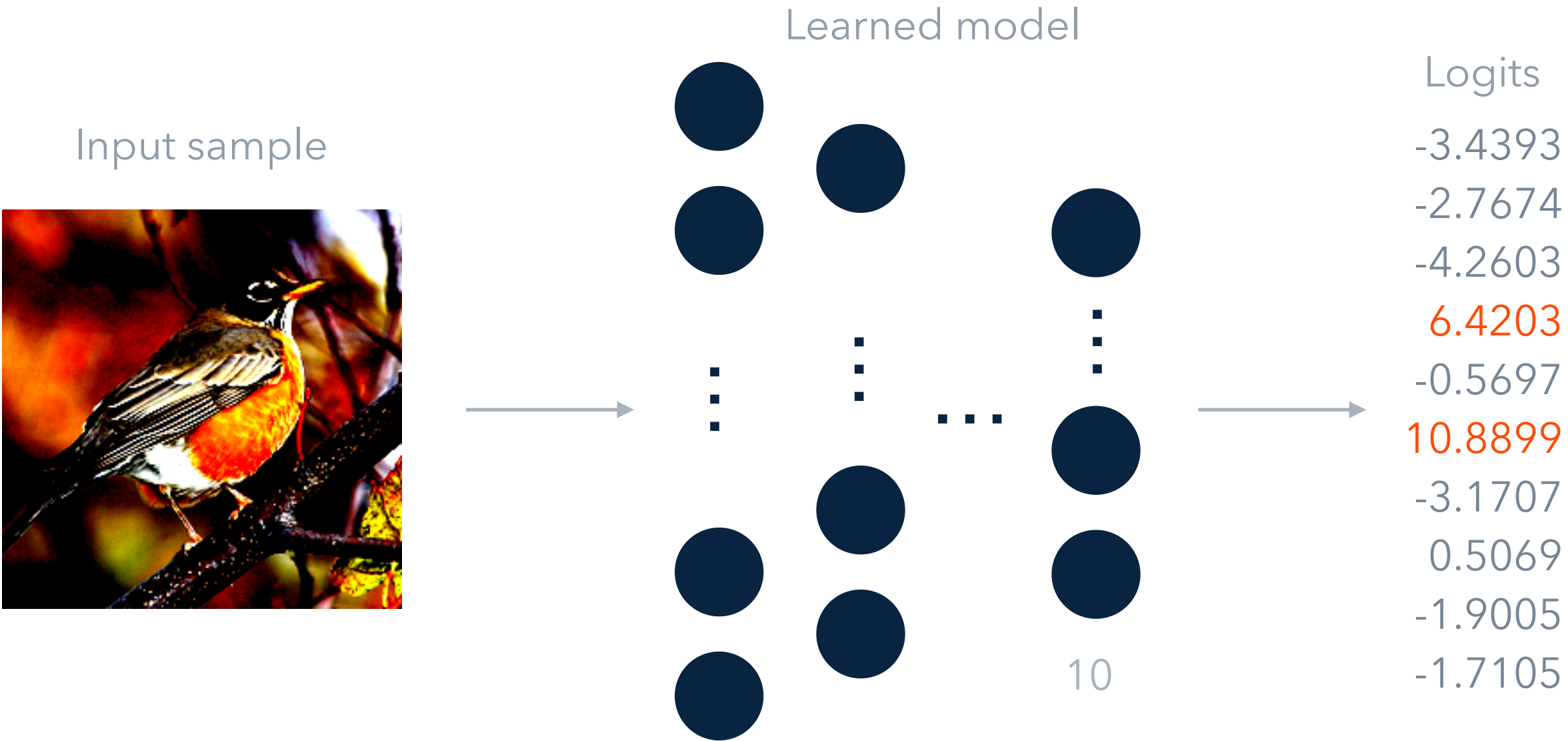
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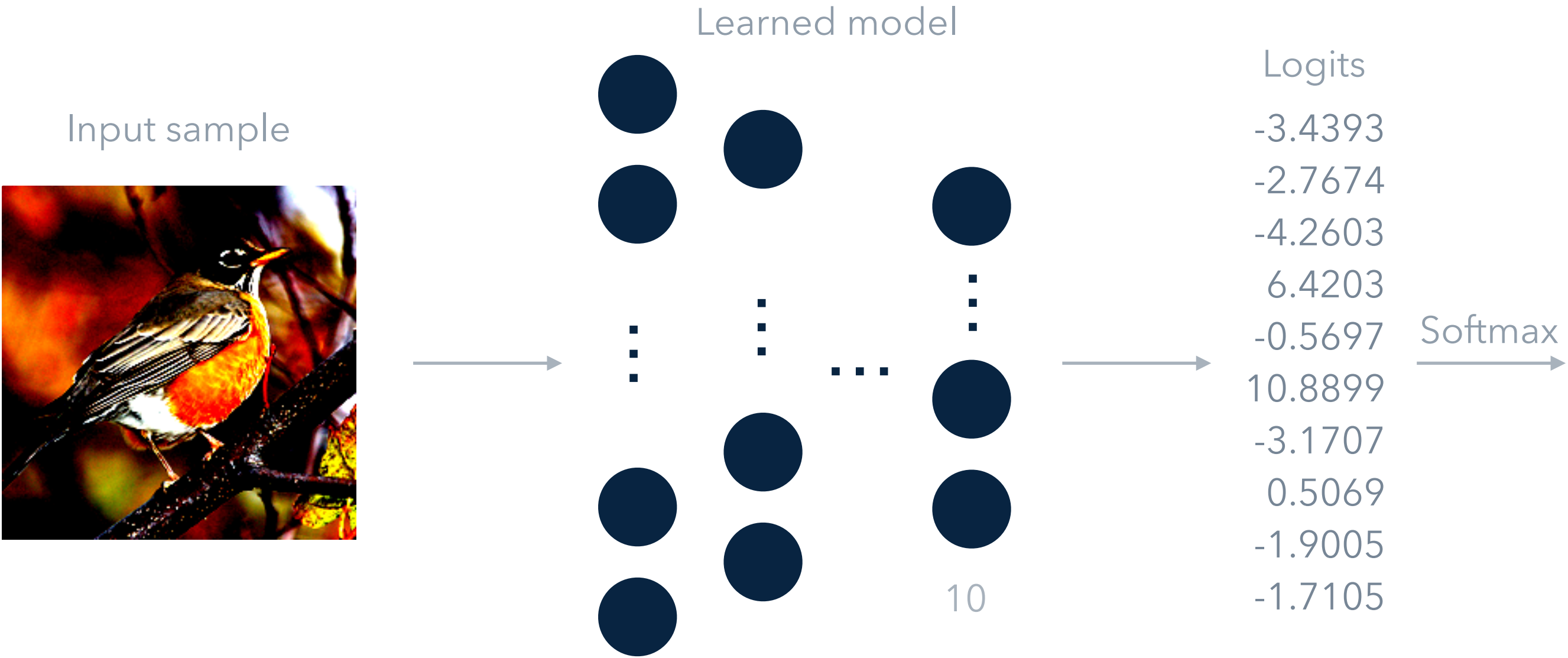
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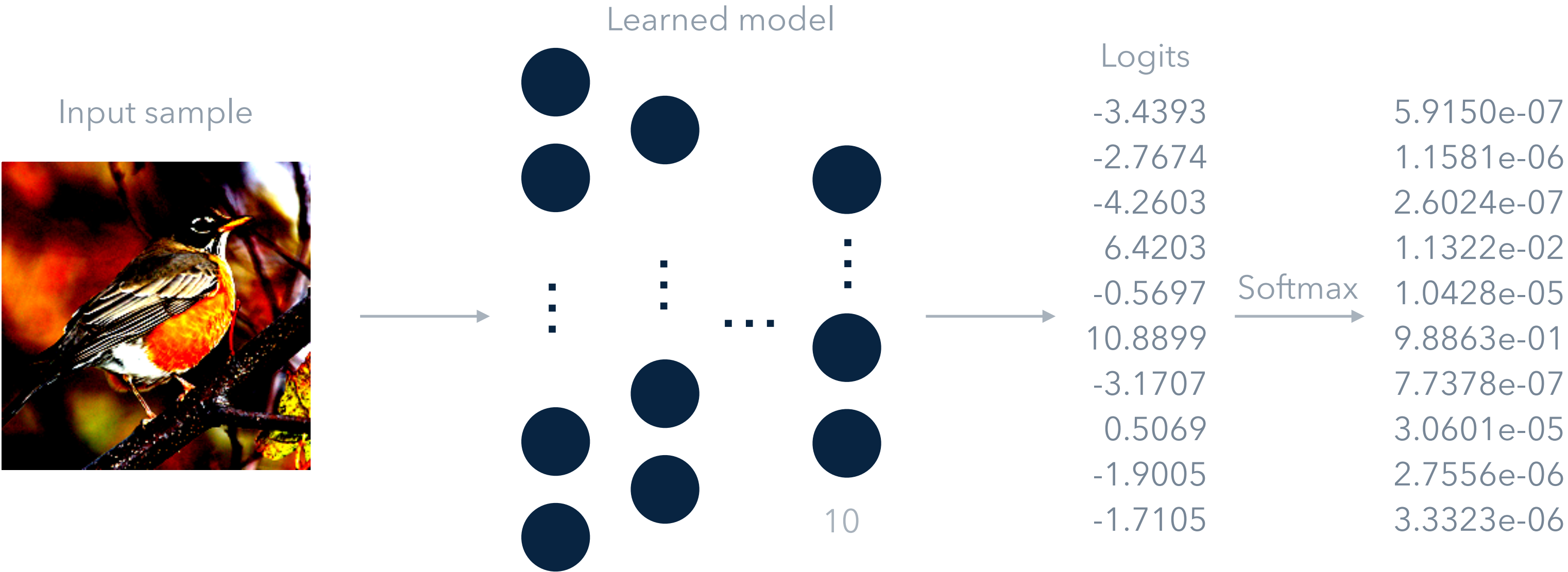
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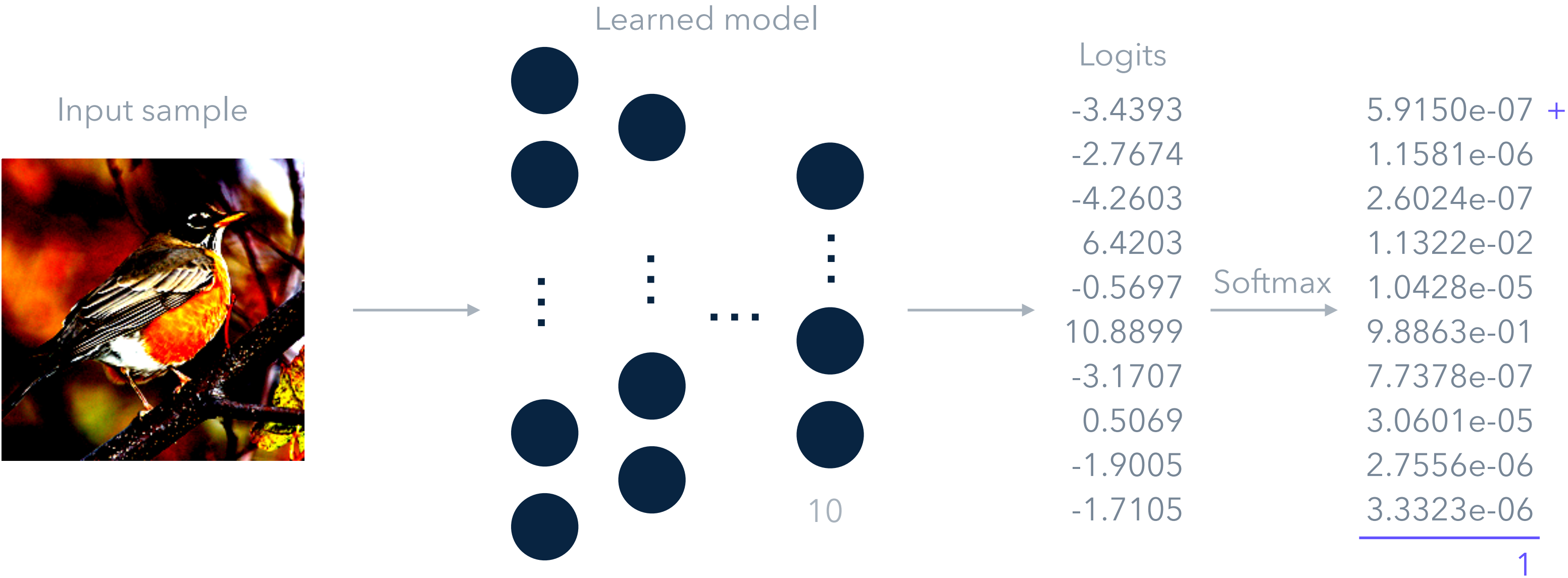
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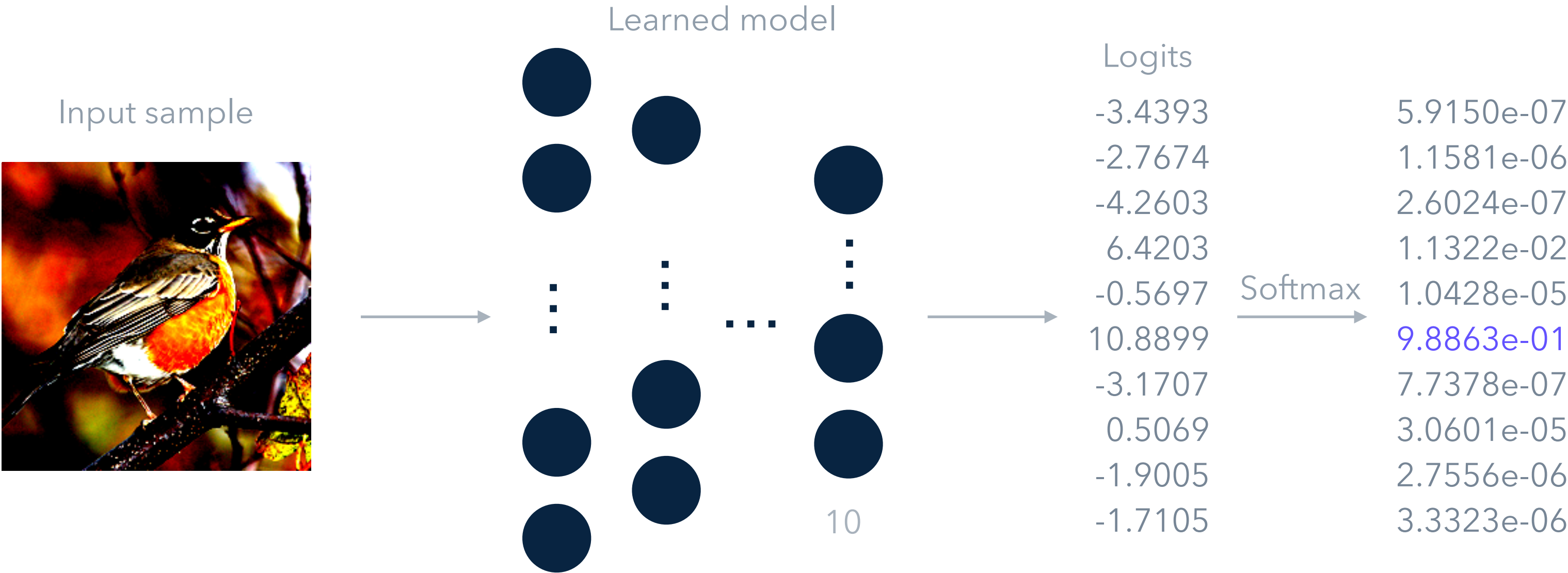
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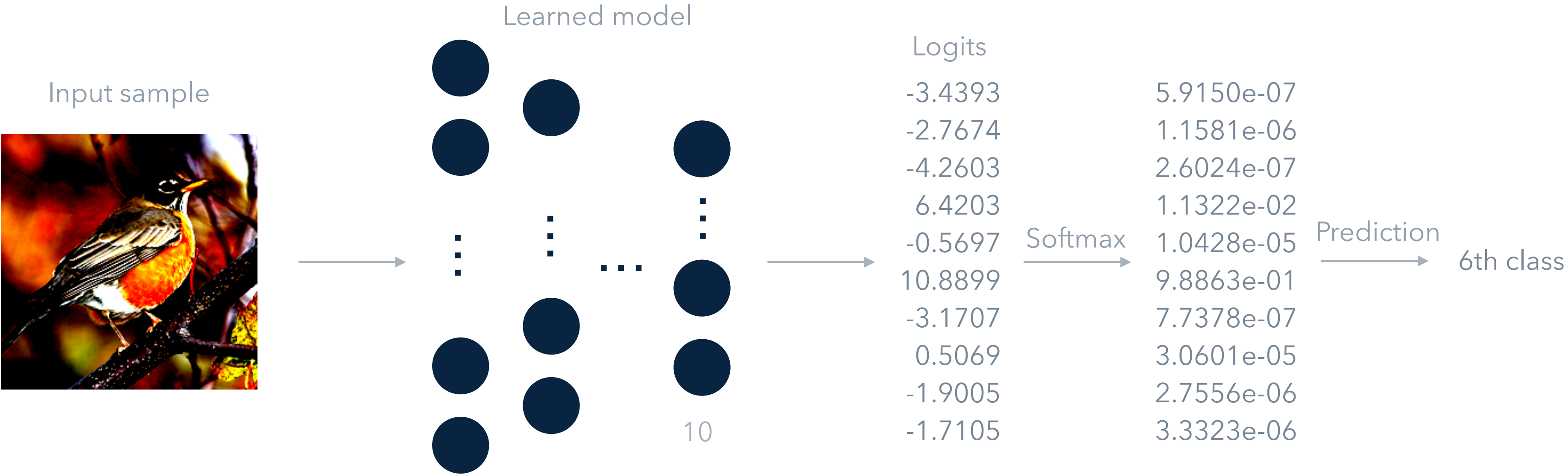
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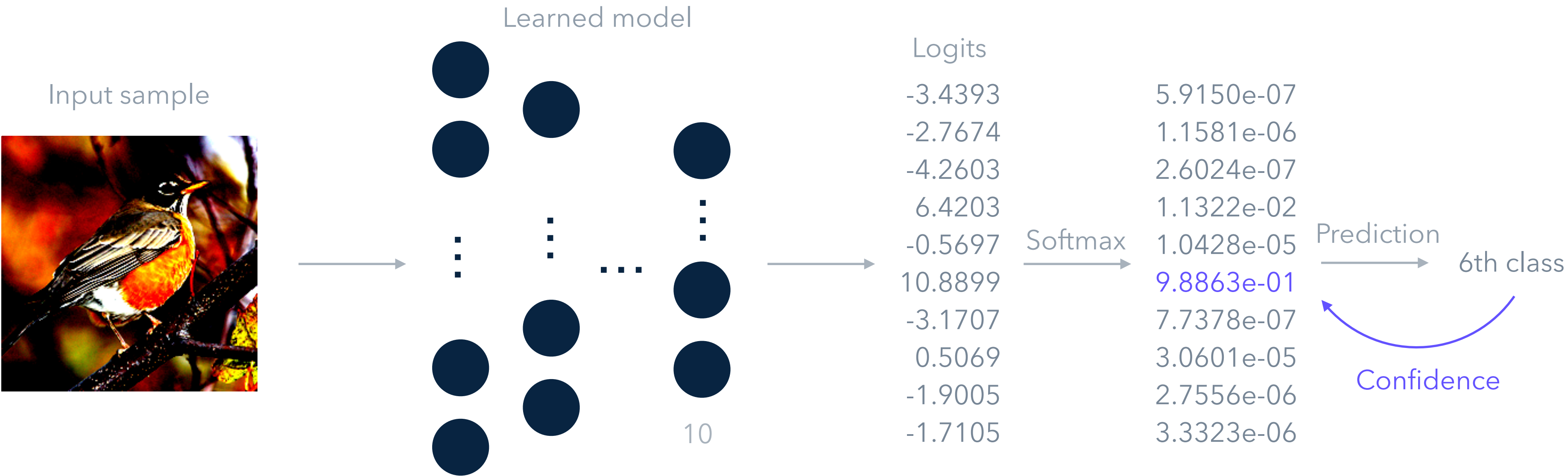
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Quick Recap



Outputs as Probabilities

Quick Recap



Reminder: How can we make sure
we don't have false negatives?

Threshold on confidence?

Calibration

- Deep learning models are very confident in their predictions

Calibration

- Deep learning models are very confident in their predictions even when the predictions are incorrect

Calibration

- Deep learning models are very confident in their predictions even when the predictions are incorrect
- This is really problematic

But why are models
overconfident?

Going Back

Let's assume

Logits		Probabilities		Target
-3.4393		5.9150e-07		0
-2.7674		1.1581e-06		0
-4.2603		2.6024e-07		0
6.4203		1.1322e-02		0
-0.5697	Softmax	1.0428e-05	Prediction	0
10.8899	→	9.8863e-01	6th class	1
-3.1707		7.7378e-07		0
0.5069		3.0601e-05		0
-1.9005		2.7556e-06		0
-1.7105		3.3323e-06		0

Cross Entropy Loss

Target	Logits		
0	-3.4393	Cross Entropy Loss	0.0114
0	-2.7674		
0	-4.2603		
0	6.4203		
0	-0.5697		
1	10.8899		
0	-3.1707		
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Cross Entropy Loss

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Cross Entropy

Loss

0.0114

Cross Entropy Loss

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Back to Calibration

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Back to Calibration

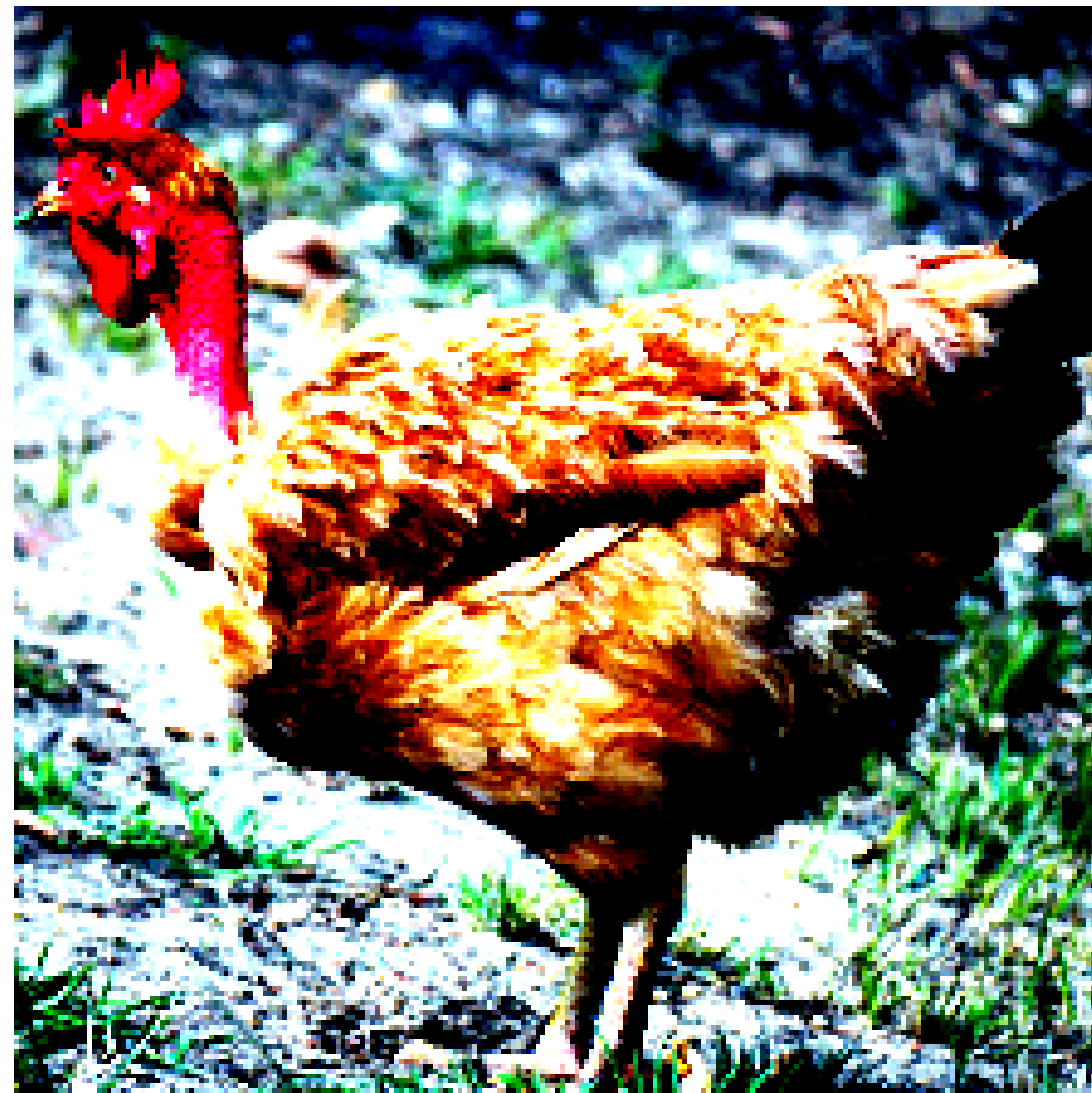
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- We are currently rewarding models for being overconfident
- There are methods that aim to make models more calibrated (e.g. temperature scaling, mixup) but they are still limited
- This an open research question

Reminder: Model Qualities

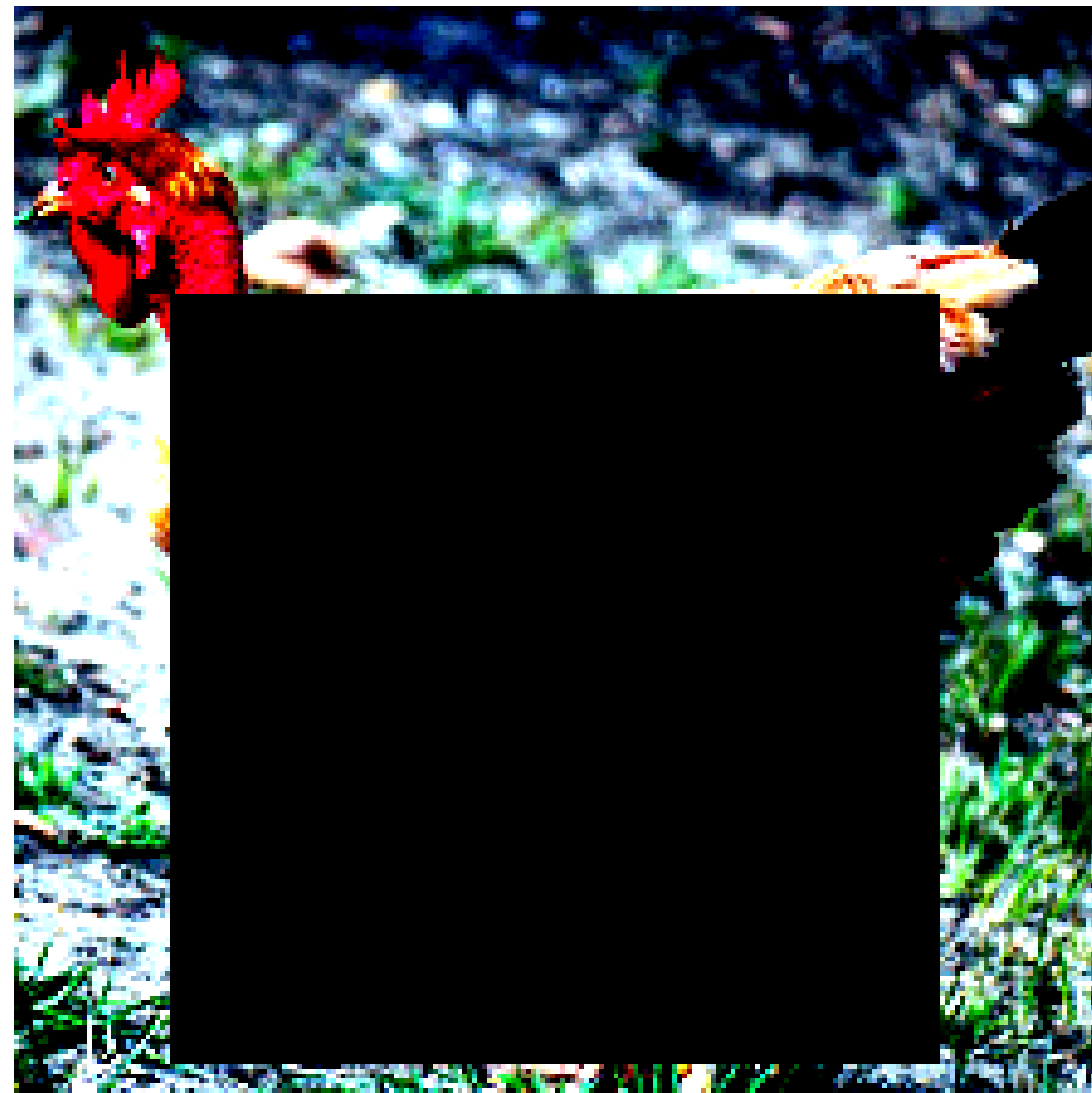
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 3. Expressivity

2. Robustness

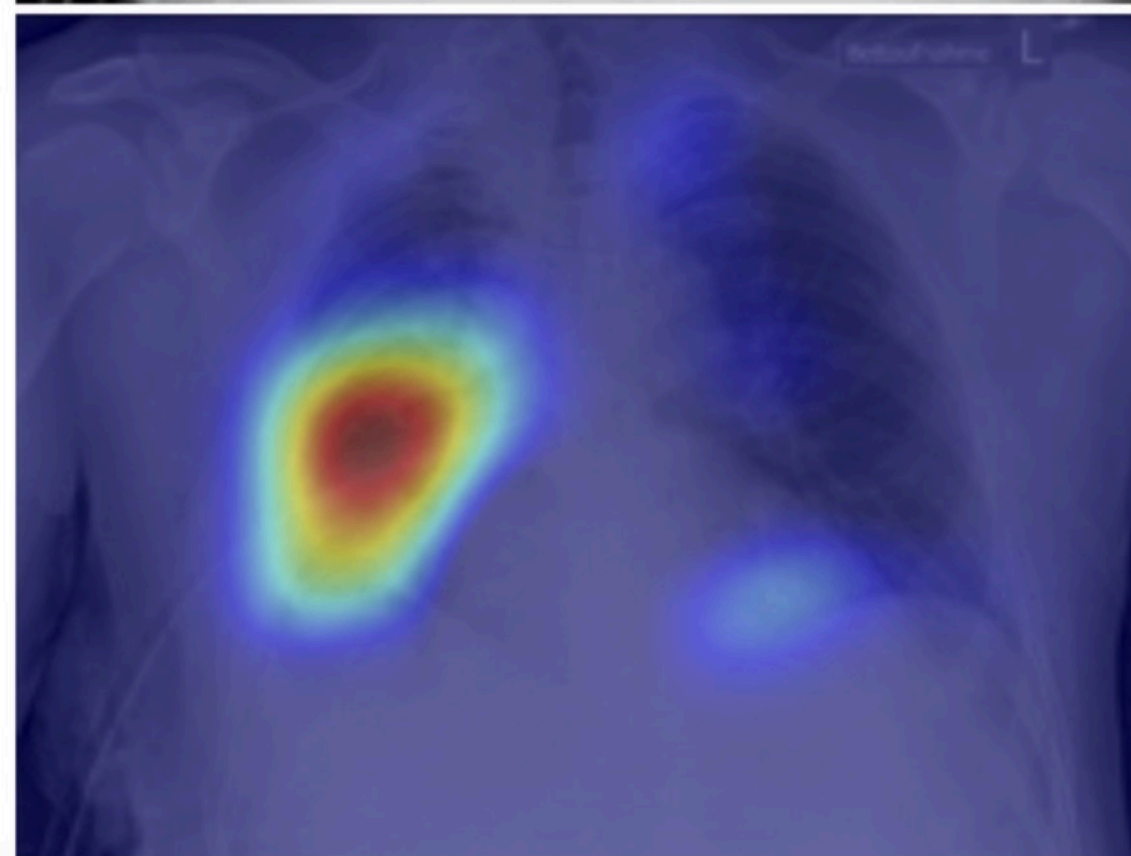
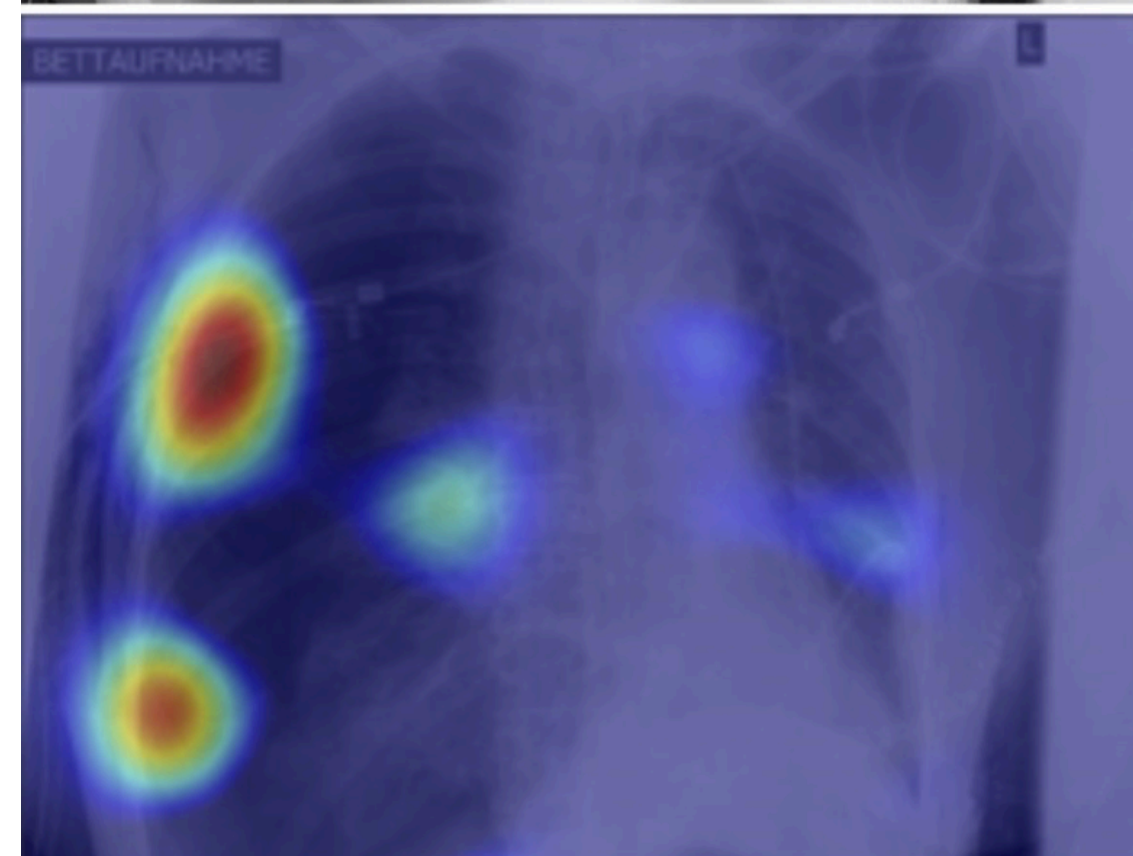
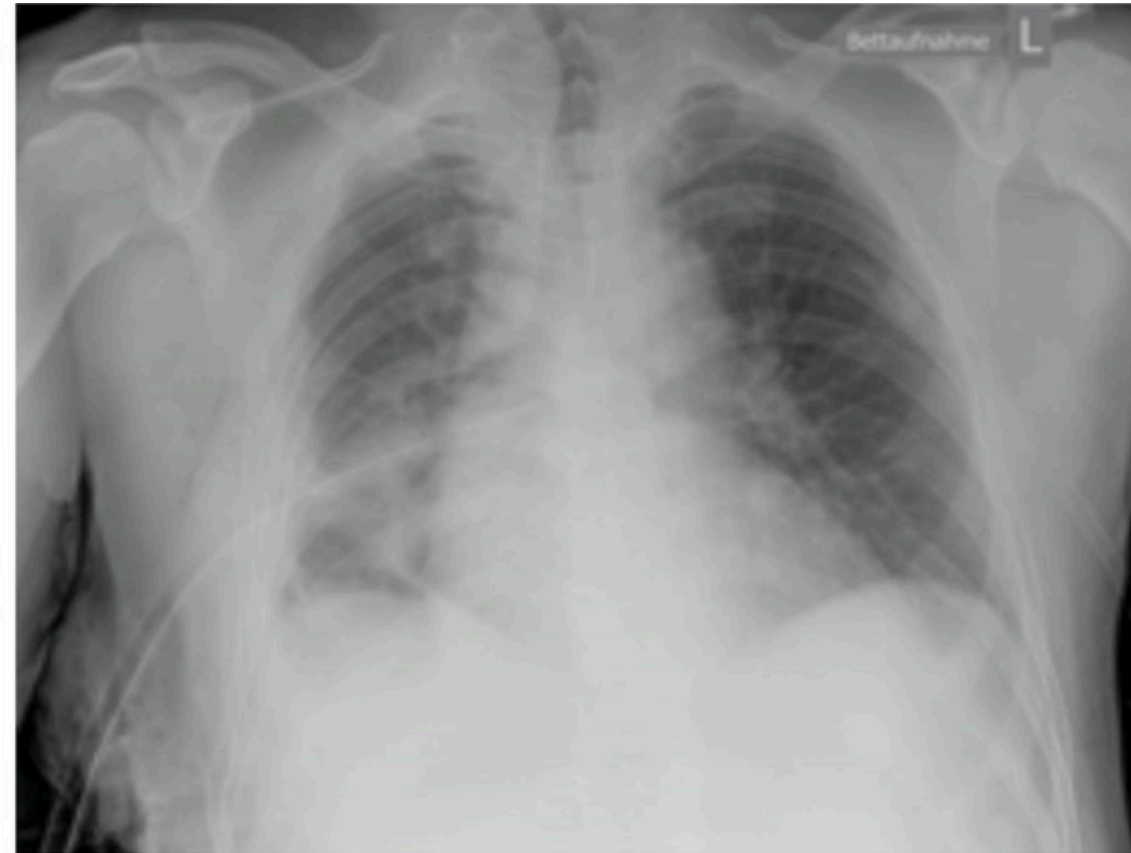
Robustness Example



Robustness Example



Going Back...



Robustness is a huge topic

Reminder: Model Qualities

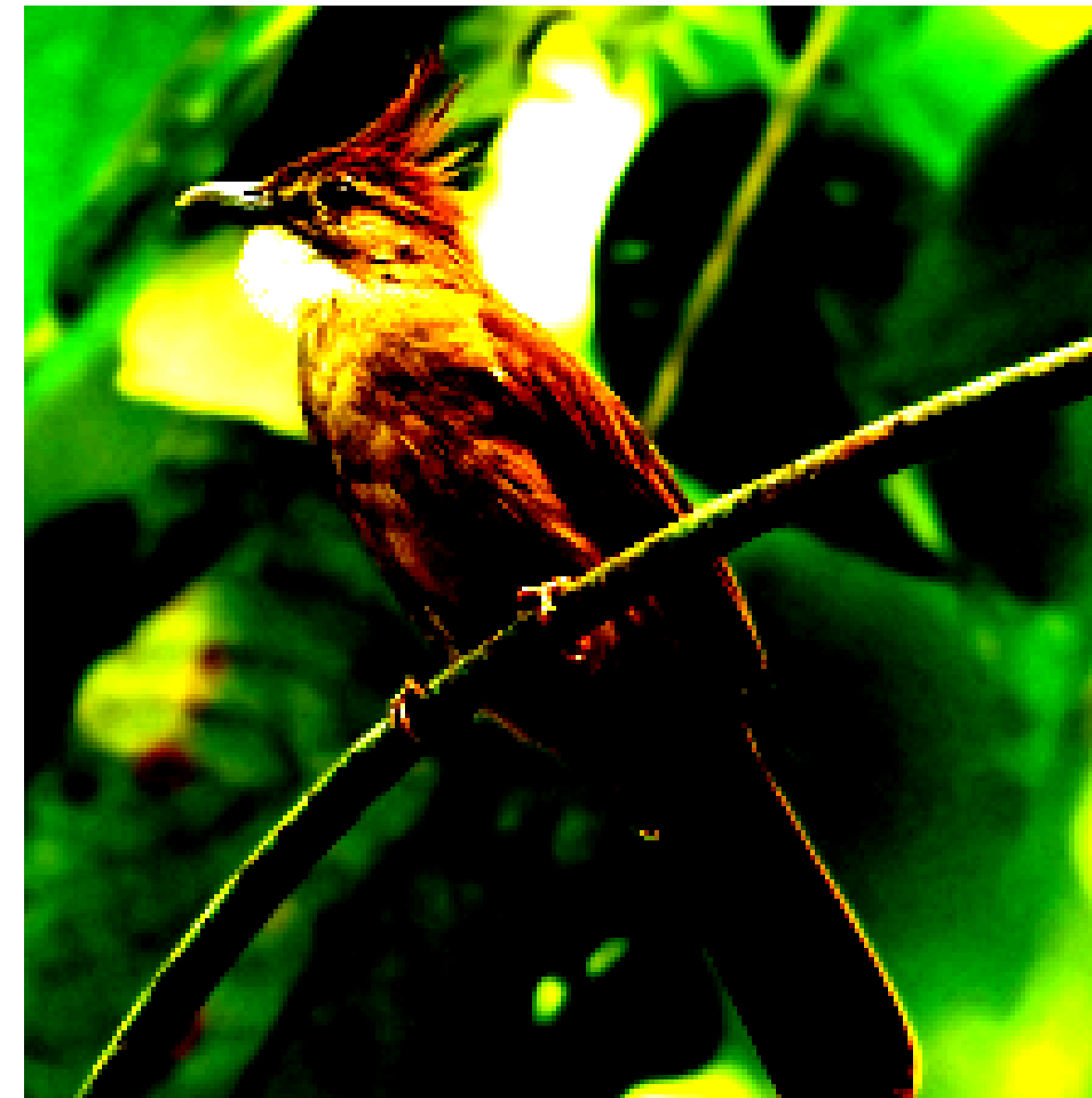
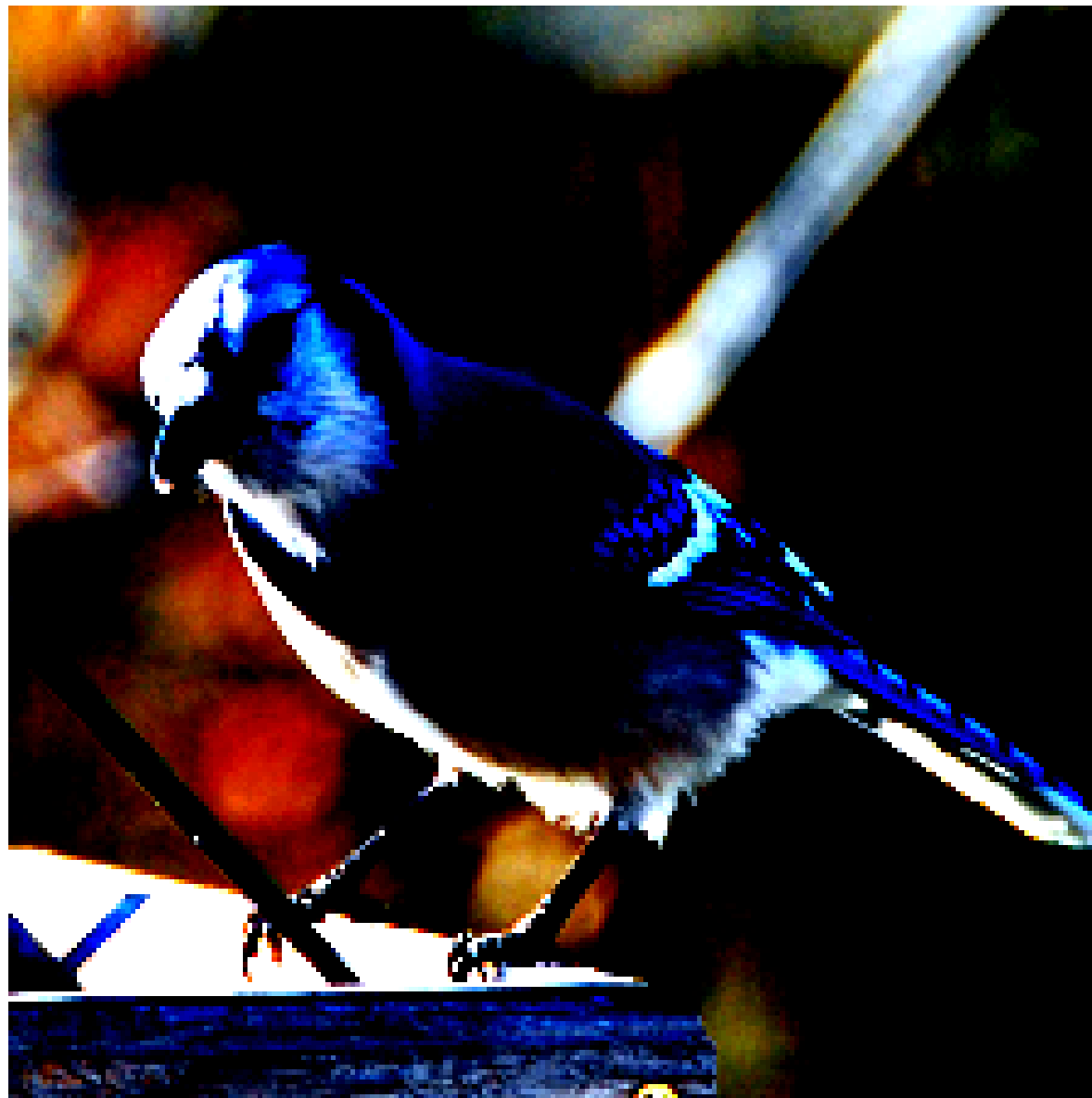
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3. Expressivity
(= Complexity, Compression)

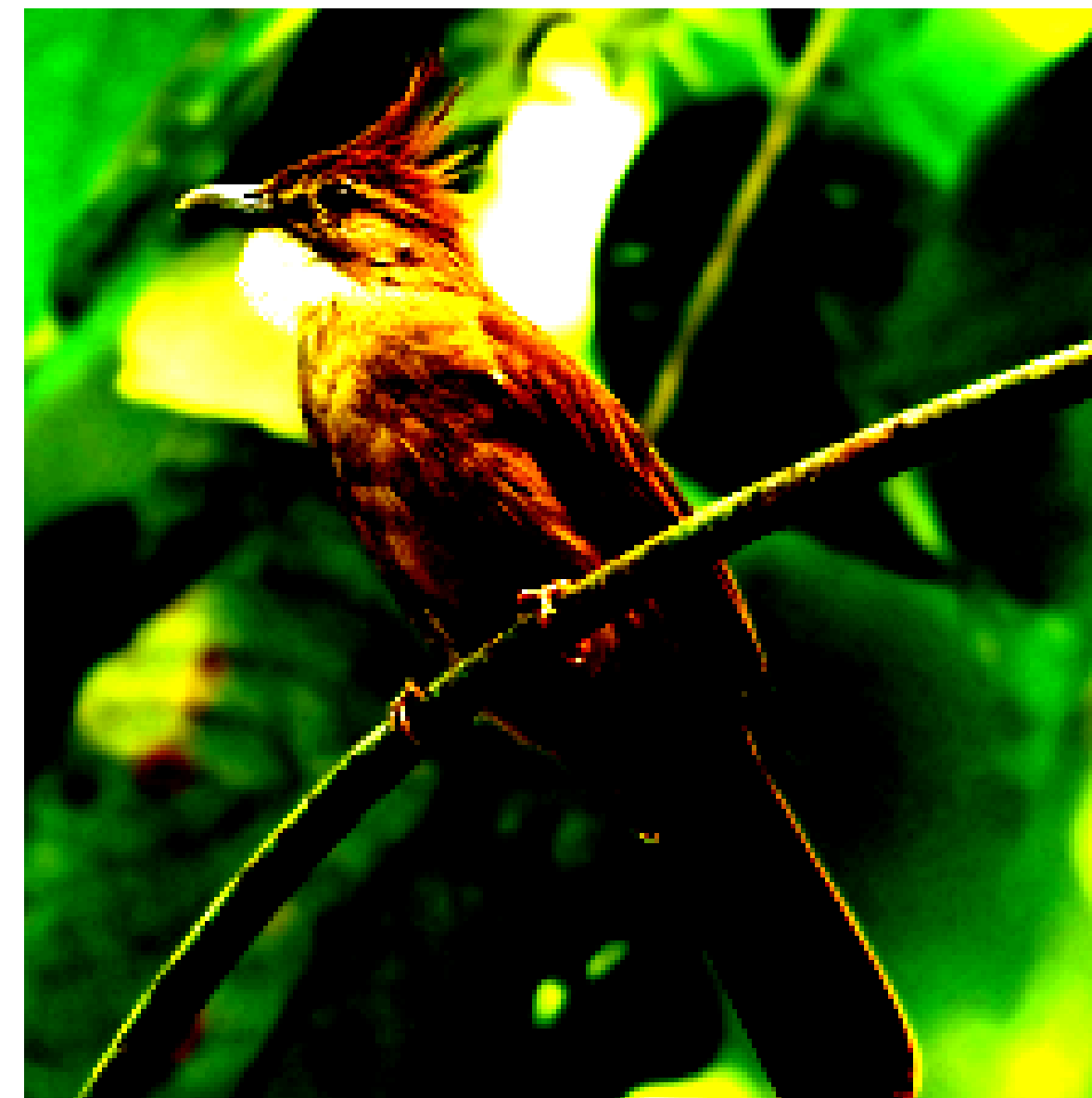
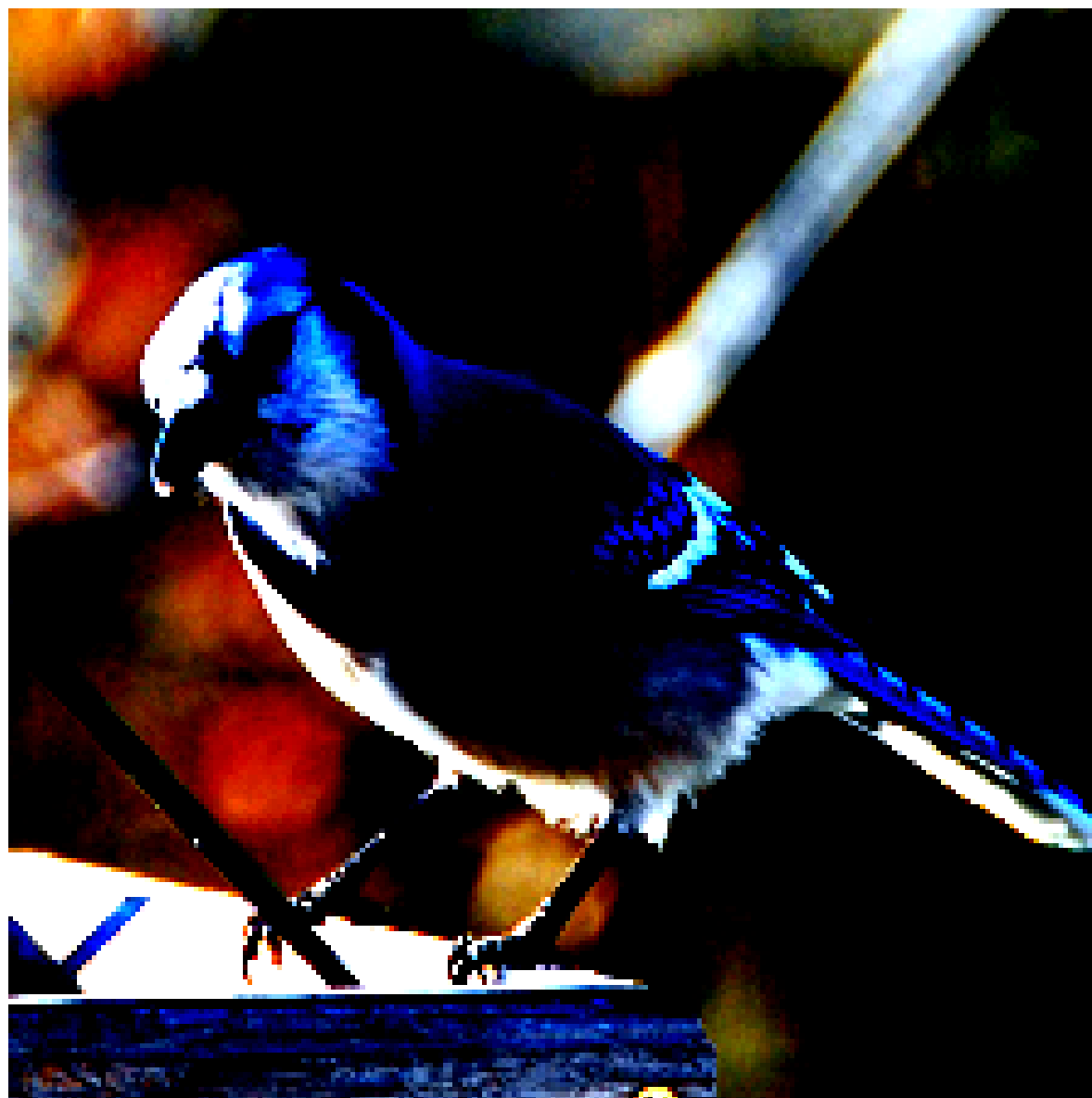
Expressivity

- Classify these two types of birds.



Expressivity

- Classify these two types of birds. Do you need to look at every single pixel?



Data is complex, but decisions
don't necessarily have to be

Complexity

- What notions of complexity did you previously learn about?

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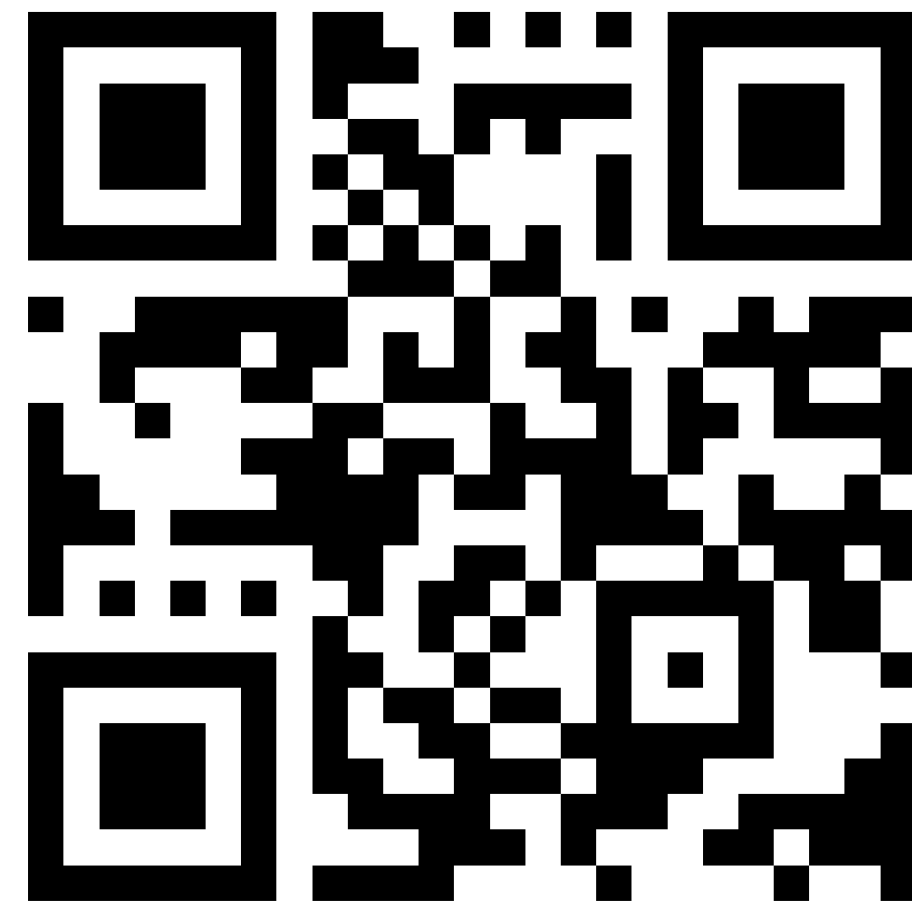
- What notions of complexity did you previously learn about? **Number of parameters?**
- In the types of models discussed so far, expressivity increases with the number of parameters
- CNN expressivity is not necessarily linear in the number of parameters
- Not suitable for comparing CNN against CNN

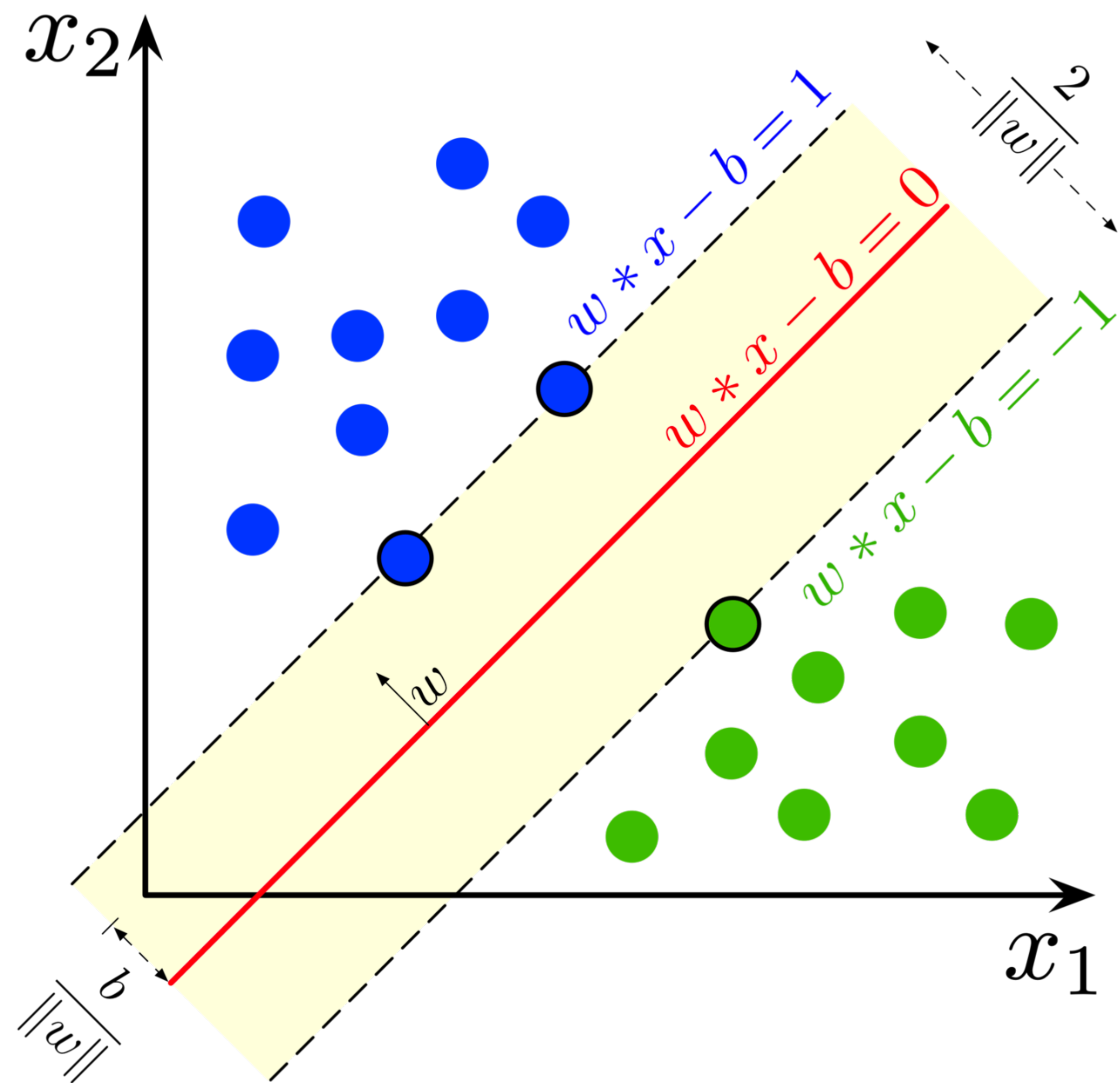


Expressivity of a learned model

- What is the expressivity of a learned model?
 - The number of parameters is not a great notion
 - What about the magnitude of the parameters (weights)?

Which of the following do you think of when you hear “weight magnitude/norm”?





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 - What about the magnitude of the parameters (weights) or the margin?
 - More advanced topic: They don't correlate well with generalisation yet!
- We don't have good ways of measuring expressivity of modern ML models

Final Reminder: Model Qualities

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 1. Calibration
 2. Robustness
 3. Expressivity

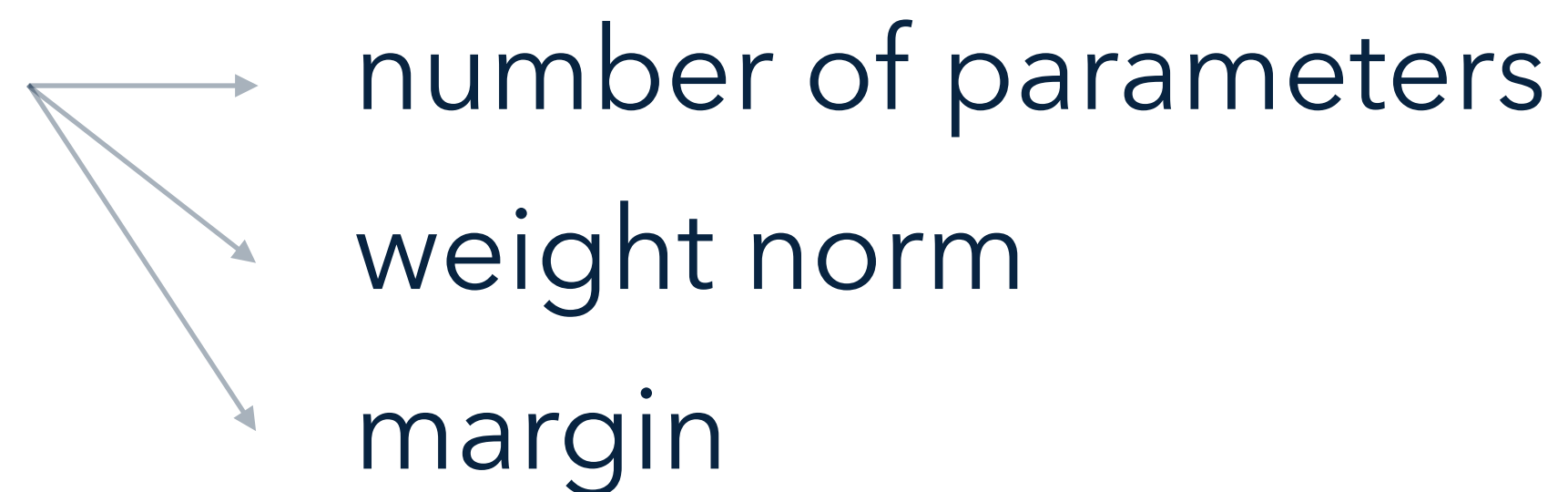
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- Model quality is not just accuracy!
- It's difficult to make sure the models we create are safe
- Researchers agree on some desirable properties but we don't have good ways of measuring all of them in all scenarios
- Making machine learning trustworthy is very much work in progress

Questions?

A.Marcu@soton.ac.uk

Excited about machine learning?

Read on

Want to know more?

Why not look up:

- Causality
- Simplicity bias
- Out-of-distribution generalisation
- Adversarial robustness
- Inductive bias
- Distribution shift
- Representation similarity measures
- Permutation symmetries
- Memorisation

Extra resources

Want to read more about the topics discussed today?

- Statistical learning theory:
 - A great textbook + YouTube lecture series from the same author
- Empirical theory of generalisation:
 - The paper that started this field
 - A NeurIPS (one of the biggest ML venues) competition + paper

Extra resources

- Gradient-based visualisations:
 - Grad-CAM is the best known method
 - Here is some criticism of these methods (in reality there are more issues)
- Calibration
 - Here is an iconic paper on calibration